

Project Time Management

Topic # 5A

Chapter 6 - PMBOK

Chapter 6 - Schwalbe

## Main Processes in Project Time Management:

- Identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables.
  - *An **activity** or **task** is an element of work normally found on WBS that has an expected duration, a cost, and resource requirements.*
- Identifying & documenting the relationships between project activities.
- Estimating how many & type of resources to complete the activity
- Estimating the number of work periods that are needed to complete individual activities.
- Creation of project schedule using activity sequencing, activation duration estimates & resource requirements.
- Involves managing & controlling project schedules.



Scheduling (Activities)

- A software project is accomplished or built through a number of smaller tasks.
- While some of these tasks can be completed at their own pace without affecting the project as a whole, there are certain tasks, which are very critical.
- These critical tasks have a deep impact on the project and if they fall behind schedule, the entire project is affected and will be delayed.
- It is the responsibility of the ***project manager*** to identify the critical tasks and later track their progress so that any delay is recognized and corrected immediately.
- The project manager can achieve this provided he has a schedule that will enable him to monitor and track the progress of the project.
- Such a schedule is termed as a ***software project schedule***.
- ***Generating Activity List***
- Assigning and Knowing Activity Attributes
- ***Defined outcomes***
  - Every scheduled task should have a specific outcome. Outcome could be products, also called deliverables.
- ***Milestone*** – an event signifying the completion of a major project deliverable.

## Activity Sequencing

- A dependency or relationship relates to the sequencing of project activities or tasks.
- Basic Reasons for Dependencies:
  - Mandatory Dependencies
  - Discretionary Dependencies: Defined by project teams
  - External dependencies: Involves relationship b/w project & non project activities




## Lead & Lag Time

- ***Lead Time***: When first activity is still running the second activity starts, the remaining time left for first activity is lead time.
- ***Lag Time***: When there is a delay or wait period for the second activity to start even after the finish time of the predecessor it is called lag time.

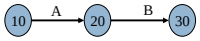


# Network Diagram

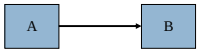
- A task network also called an ***activity network*** is a graphical representation of the task flow of the project.
- Helps to identify task sequences, interdependencies & parallel activities
- Identify critical path tasks/activities, tasks that must be completed on schedule if the project as a whole has to be completed on schedule
- Predecessor v/s Successor Activity
- Network Diagrams
  - Activity on Node (AON)
  - Activity on Arrow (AOA)

# AON V/S AOA

- ***Activity on Arrow***




  - Arrow represents activities, nodes represents events
- ***Activity on Nodes***



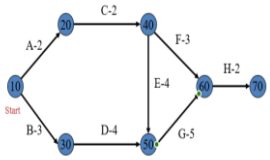

  - Nodes represents activities, arrow represents order of activities.

# Critical Path & Slack

- ***Critical path*** – the sequence of dependent tasks that determines the earliest possible completion date of the project.
- ***Slack or Float time*** – the amount of delay that can be tolerated between the starting time and completion time of a task without causing a delay in the completion date of the entire project.

# Example

| Code | Time (weeks) | Predecessor |
|---|---|---|
| A | 2 | - |
| B | 3 | - |
| C | 2 | A |
| D | 4 | B |
| E | 4 | C |
| F | 3 | C |
| G | 5 | D,E |
| H | 2 | F,G |




# Network Models & Critical Path

- Find All Paths & Duration (ACFH, ACEGH, BDGH)
- Critical Path,
- Project Completion Time,
- Critical Activities
- Project completion 15 weeks

# Start Times & Finish Times

- Earliest Start Time (EST) & Earliest Finish Time (EFT)
  - Assume all the activities start as soon as all predecessors complete
  - Early start time should be calculated considering the last activity early finish time whereas finish time can be calculated by adding the duration of the activity to the current activity start time
- Late Start Time (LST) & Late Finish Time (LFT)
- LST for an activity is minimum of LFT minus the duration, for all of its succeeding activities
- –LFT = LST of all successor activity – 1



## Forward Pass

- **Step 1:** Make a forward pass through the network as follows: For each activity *i* beginning at the Start node, compute:
  - **Earliest Start Time (ES)** = the maximum of the earliest finish times of all activities immediately preceding activity . (This is 0 for an activity with no predecessors.). This is the earliest time an activity can begin without violation of immediate predecessor requirements.
    **Earliest Start Time (ES)** = the **earliest time** that an **activity** can **begin**
  - **Earliest Finish Time (EF)** = This represent the earliest time at which an activity can end.
    **Earliest Finish Time (EF)** = the **earliest time** that an **activity** can **be completed**

The project completion time is the maximum of the Earliest Finish Times at the Finish node.

## Backward Pass

**Step 2:** Make a backwards pass through the network as follows: Move sequentially backwards from the Finish node to the Start node. At a given node, *j*, consider all activities ending at node *j*. For each of these activities, (*i*,*j*), compute:

- **Latest Finish Time (LF)** = the minimum of the latest start times beginning at node. This is the latest time an activity can end without delaying the entire project.
  he **earliest time** than an **activity** can be **completed without *lengthening* the minimum project duration**
- **Latest Start Time (LS)** = This is the latest time an activity can begin without delaying the entire project.
  the **latest time** that an **activity** can **begin without *lengthening* the minimum project duration**

## Project Duration

**Duration of the project** is the **difference** between the ***maximum*** **value of the "latest finish time"** of projects and the ***minimum*** **value of the "earliest start time"** of projects

`Project duration = max(LF) - min(ES)`

## Total Float & Free Float

- ***Total Slack or Float*** is the amount of time a task can be delayed without delaying the project's finish date.
  - ES- LS or EF-LF

A critical path is a path of activities, from the Start node to the Finish node, with 0 slack times.

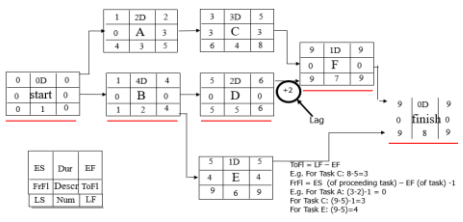
**Critical activity** = an **activity** in the project with **slack = 0** i.e.: **critical activities** have **no slack**, when such project is **delayed**, the **duration of the project will *increase***

- ***Free Slack*** is the amount of time that a task can be delayed without delaying the earliest start time of any successor task.
  - Calculated by difference between early start of next and early finish of current

## Example

| Activity | Estimated Duration (Days) | Constraints |
|---|---|---|
| A | 2 | Can start at any time |
| B | 4 | Can start at any time |
| C | 3 | Cannot start until A has ended |
| D | 2 | Cannot start until B has ended |
| E | 1 | Cannot start until B has ended |
| F | 1 | Cannot start until C has ended and 2 days after D has ended |

## Solution Using Activity on Node

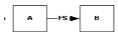


## PDM (Precedence Diagramming Method) OR AON

- Uses boxes or rectangles referred as nodes to represent activities.
- Arrows show relationships between activities.
- This technique is also called Activity-On-Node (AON)
- PDM includes four types of dependencies or logical relationship.

## Activity Relationships

- **Finish to Start:** Activity A must be completed before Activity B starts



- **Start to Start:** Task A cannot Start until Task B starts



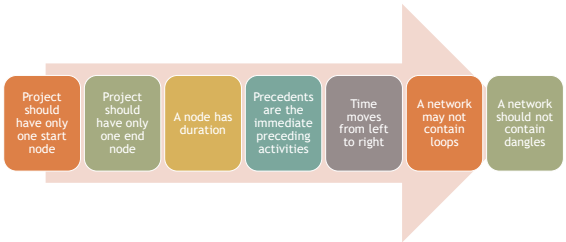
- **Finish to Finish:** Task B cannot finish until Task A finishes



- **Start to Finish:** Task B cannot finish until Task A starts (Rarely used)

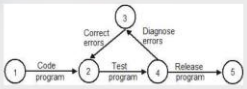


## Recap: Constructing Network Diagrams

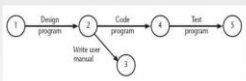



## Activity on Arrows Networks: NOTS

- A network may not contain loop

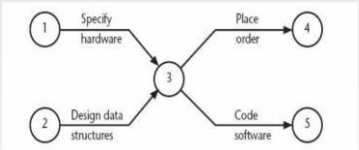



A network may not contain dangle

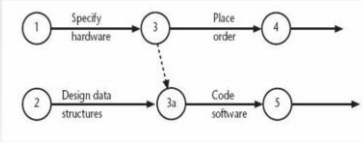



## AOA: Using Dummies

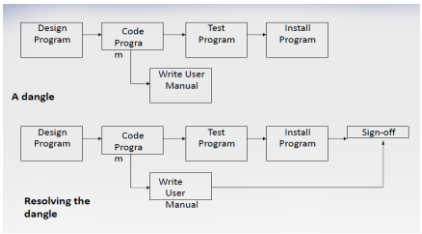
A logical error occurs when two paths ('specify hardware' and 'design data structure') within a network have a common event although they are independent




Separating the two independent paths and introducing a dummy activity to link the completion of 'specify hardware' to the start of the activity 'code software'

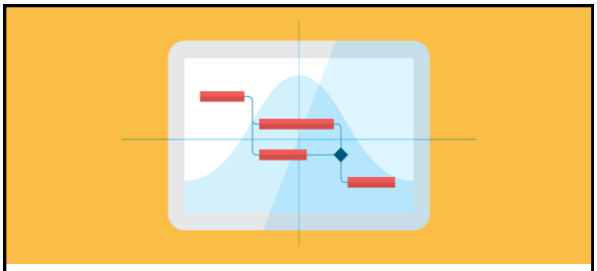


## Network: resolving dangles



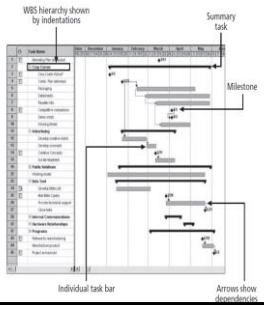

## Schedule Development – Tools & Techniques

| | |
|---|---|
| Gantt Chart | • A common tool for displaying project schedule information. |
| Critical Path Method | • **(CPM)** is a very important tool for developing and controlling project schedules. |
| PERT | • **PERT** analysis is a means for considering schedule risk on projects. |



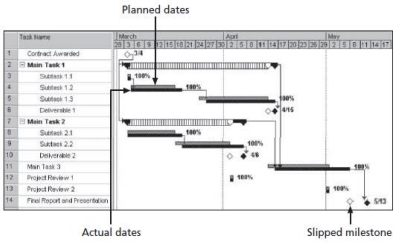
## Gantt Charts

- **Gantt charts** provide a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in a calendar format.
- Gantt chart don't show relationships among activities as network diagrams do.
- Activities on left, horizontal bars under calendar showing start and end of the activities.
- Gantt charts can also be used to track the progress of the project by showing actual schedule information.




## Gantt Charts

- Sample tracking Gantt Chart




## CRITICAL PATH METHOD



- It is a project network analysis technique used to predict total project duration.
- The network model developed helps to determine all paths, critical paths, total and free floats, early start, early finish, late start & late finish.

# END OF TOPIC 5A

COMING UP!!!!!!
HR - Project Team Organization
Project Cost Management
Project Control



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